OCR and Machine Translation Pipeline for Low-Resource Sámi Languages

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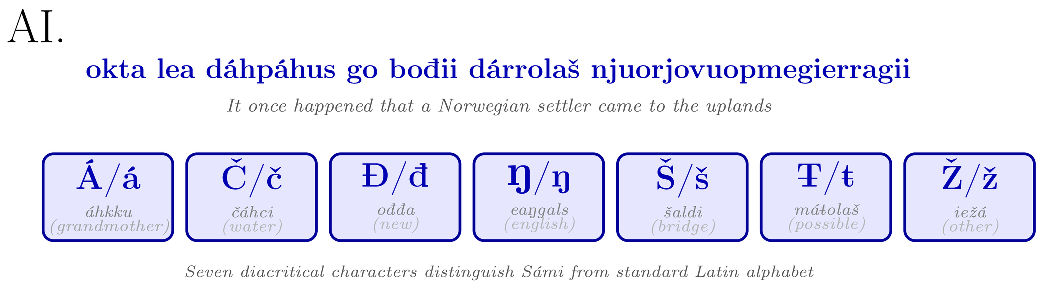
1. Background & Motivation

Sámi languages are a group of nine low-resource Uralic languages spoken across Northern Europe. Northern Sámi, the most widely spoken, has approximately 25,000 speakers and is classified as endangered by UNESCO.

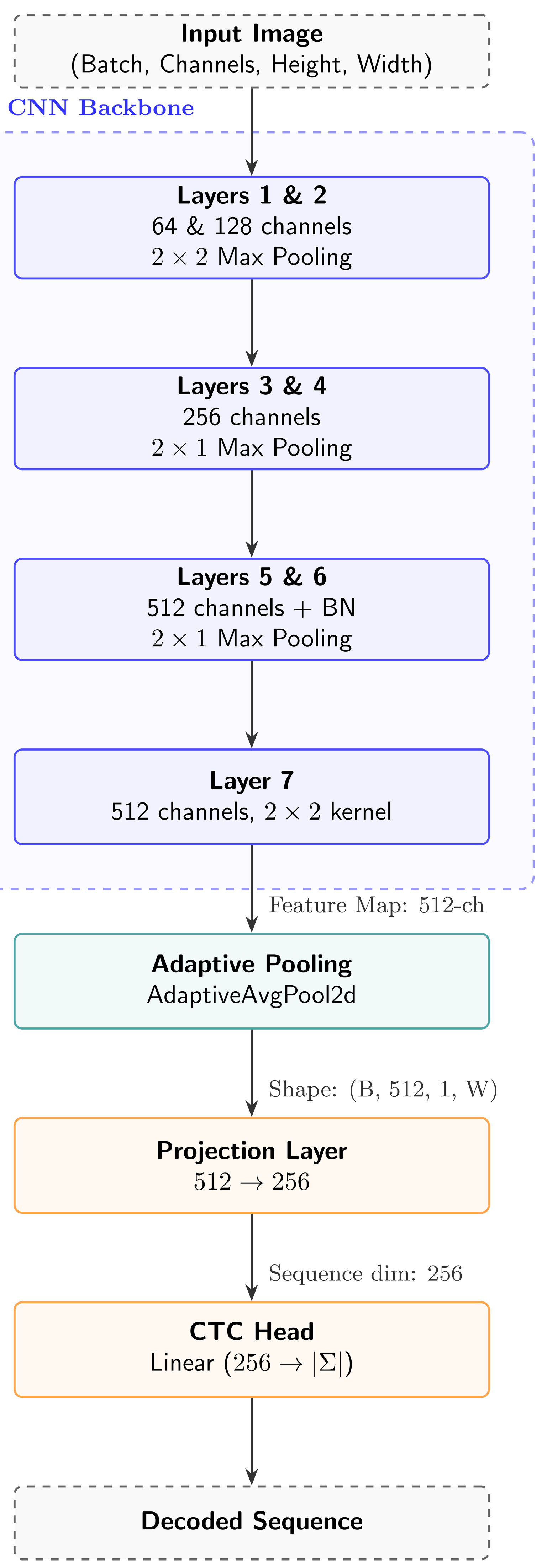
Challenges in Optical Character Recognition (OCR) for North Sámi range from scarce training data to complex orthographic features, making it a difficult task to streamline the digitalization efforts. Additionally, the language characteristics of Sámi lead to a large number of inflected word forms, further complicating accurate recognition. While state-of-the-art models like TrOCR achieve high accuracy (91.56%), their computational demands (~350MB, ~12.41 s/img inference on CPU) make them ill-fitted for deployment in resource-constrained environments.

Objective was to develop and benchmark a lightweight CNN-CTC OCR model for Northern Sámi, evaluating its performance against the TrOCR transformer-based models and in different contexts such as in OCR→MT translation pipeline, out-of-domain generalization, linguistically motivated optimisation and applicability in sustainable AI.



(a) Example sentence from Johan Turi's *Muitalus sámiid birra* (1910). (b) Seven Sámi-specific diacritics that distinguish the orthography.

Architecture



CNN-CTC: 32×800 px input → CNN Backbone → Adaptive Pooling → Projection Layer → CTC head → Decoded Sequence.

2. Data, Methods & Metrics

Training data was Språkbanken synthetic Sámi OCR dataset (National Library of Norway), ~307k line images (276k training + 31k hyperparameter tuning), 32×800px.

Evaluation data used 1048 samples: 1000 in-domain synthetic lines (Språkbanken validation split) + 48 parallel Northern Sámi–English sentences from Turi's 1910 "An Account of Sámi".

Pipeline - a lightweight, modular OCR→MT:
1. Image preprocessing (grayscale, 32×800 px)
2. 7-layer CNN backbone ($64 \to 128 \to 256^2 \to 512^3$ ch.)
3. Adaptive pool + linear projection to 256 dimensions
4. CTC decoder over a 395-symbol vocabulary
5. Translation via TartuNLP API

Training. AdamW (lr 10^{-4}), batch 32, 100 epochs with early stopping, CTC loss, trained on a single NVIDIA V100. Inference benchmark on AMD Ryzen 5 PRO CPU.

Negative result. Pretrained VGG-16/19 and ResNet-50 backbones did not converge.

Metrics Used. OCR: Character Error Rate (CER), Word Error Rate (WER); Word Acc. = 100−WER, Char Acc. measured directly (≈ 100−CER).

3.1 Results: Benchmark

Key result: CNN-CTC achieves 87.72% accuracy (12.24% CER) with the fastest inference and smallest model.

Model	Char. Acc. (%)	Word Acc. (%)	s/img	Size
trocr_smi_synth*	**91.56**	**71.84**	12.41	350MB
ctc_simple (ours)	87.72	62.65	**0.033**	**23MB**
trocr_smi_pred_synth*	84.15	54.24	12.08	350MB
trocr_smi	75.88	36.50	16.23	350MB

*`smi`=Sámi data, `pred`=+auto-transcribed, `synth`=+synthetic pre-training.

Generalization (in-domain vs. out-of-domain)

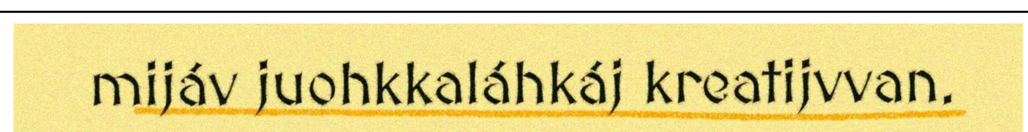
Key result: TrOCR degrades by 4.84 pp, while CNN-CTC remains stable (12.24% → 12.21%).

Model	HF CER (%)	Book CER (%)	Δ (pp)
trocr_smi_synth	**8.33**	13.17	+4.84
ctc_simple (ours)	12.24	12.21	**-0.03**
trocr_smi_pred_synth	16.09	**11.00**	-5.09

HF = in-domain, Book = historical out-of-domain.

Example predictions (ground truth vs. prediction)

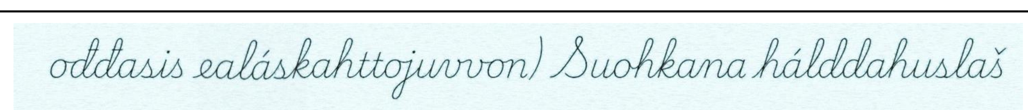
Correct Recognition (CER = 0%)



Ground Truth: *mijáv juohkkaláhkáj kreatijvvan.*

Prediction: *mijáv juohkkaláhkáj kreatijvvan.*

Diacritical Errors (CER = 8.3%)



Ground Truth: *ođđasis ealáskahttojuvvon) Suohkana hálddahuslaš*

Prediction: *oddasis ealáskahttojuvvon) Duohkama hálddahuslaš*

Uppercase Text Errors (CER = 29.6%)

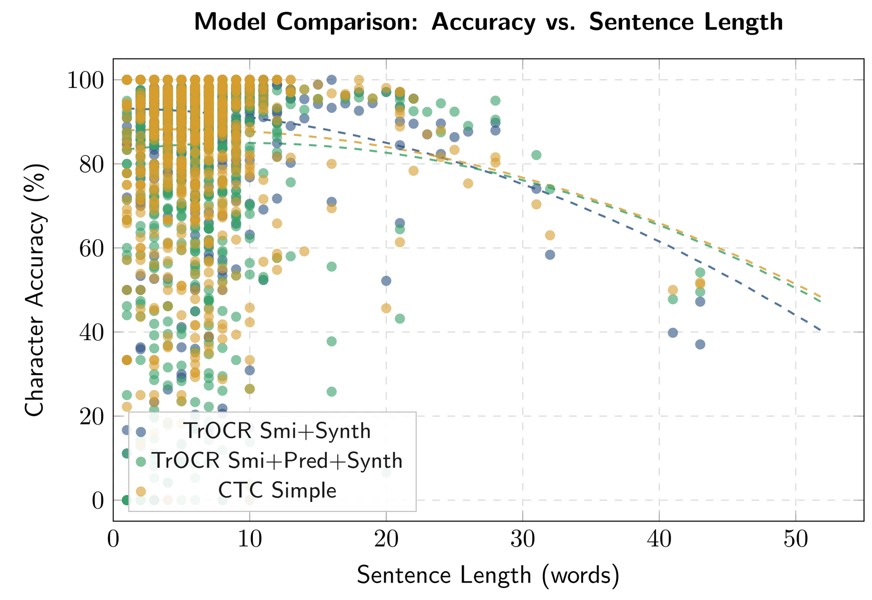
Ground Truth: *FYLKKAGIELDDAIGUIN. FINNMÁRKKU FYLKKAGIELDDAS LEA ÁIGUMUŠŠIEHTADUS SÁMI*

Prediction: *FYLKKAGIELDDAIGUI0. F1O0mÁRKKU FYLKKAGIELDDAs LEA ÁIGUmUŠe*

3.2 Results: Other Findings

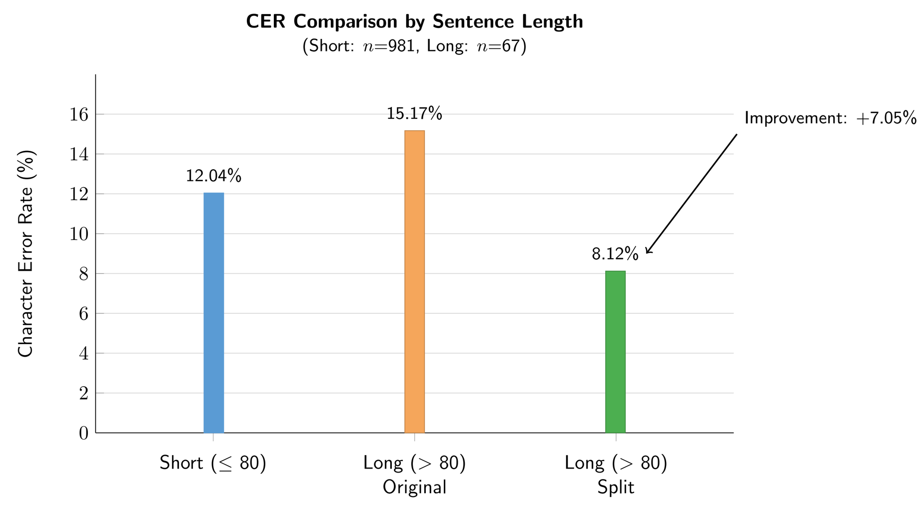
Character accuracy vs. sentence length.

The character accuracy drops for longer sentences due to the fixed 32×800 px input.



Conjunction-based splitting on long sentences

End-to-end OCR→MT: On 48 book sentences: BLEU 10.40, chrF 33.85, TER 76.30.



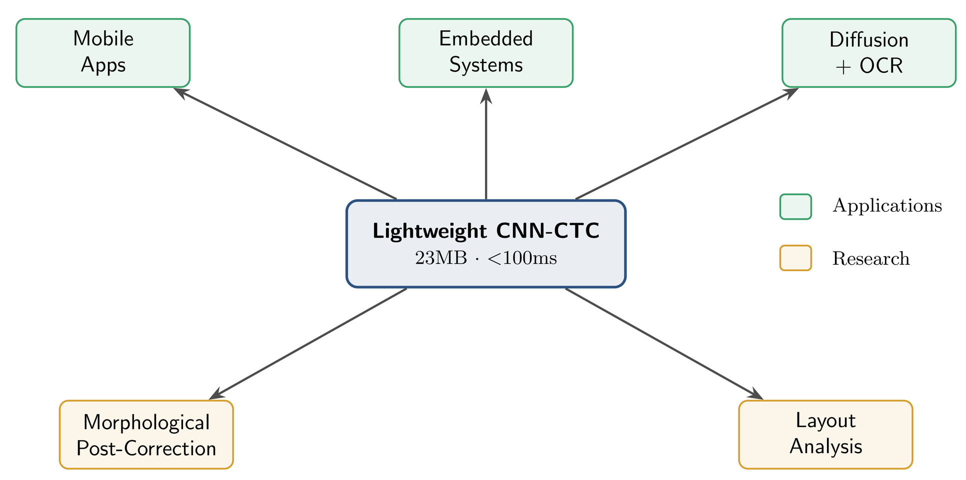
Linguistically motivated preprocessing splitting at Sámi conjunctions (*ja*, *muhto*, *go*, ...) cuts long-sentence CER from 15.17% to 8.12%.

4. Conclusion & Future Work

Key Outcomes. The CNN-CTC achieves 87.72% character accuracy. The evaluation accuracy barely moves between synthetic and historical book text (12.24% → 12.21%). Yields an estimated throughput of ~109,000 img/h. The OCR→MT pipeline achieves BLEU 10.40 on the 48 parallel sentences and conjunction-based splitting recovers ~7 pp CER on long sentences.

Implications. The CNN-CTC model trails the best TrOCR by only 3.84 pp character accuracy while being roughly 15× smaller and 380× faster, making it worthwhile for resource-constrained deployment. The in and out of domain accuracy stability lacks sufficient statistical power for concrete conclusions but it might suggest better lightweight generalisation. The MT quality indicates the need for improved English translation models for Sámi. Conjunction-based splitting proved to be a valid preprocessing proof-of-concept.

Future Work: Use real scanned documents, layout analysis for full-page processing, morphological post-correction, mobile / edge deployment for field digitization



Future work: real scans, layout analysis, morphological post-correction, mobile deployment.

References & Acknowledgments

Key references. [1] Enstad et al. *Comparative analysis of OCR for Sámi*, NoDaLiDa 2025.
[2] Li et al. *TrOCR*, AAAI 2023.
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[5] Turi, *Muitalus sámiid birra*, 1910.
[6] TartuNLP MT API, 2025.

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Code: `https://github.com/magwrap/north-sami-ocr`